

# INTRODUCTION TO MACHINE LEARNING

Week-5

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“Machine learning is the process by which a computer can work more accurately as it collects and learns from the data it is given.”

—Mike Roberts<sup>2</sup>



# APPLICATIONS FOR MACHINE LEARNING IN DATA SCIENCE

- Finding oil fields, gold mines, or archeological sites based on existing sites
- Finding place names or persons in text
- Identifying people based on pictures or voice recordings
- Recognizing birds based on their whistle
- Identifying profitable customers (regression and classification)
- Proactively identifying car parts that are likely to fail (regression)
- Identifying tumors and diseases
- Predicting the amount of money a person will spend on product X (regression)
- Predicting the number of eruptions of a volcano in a period (regression)
- Predicting your company's yearly revenue (regression)
- Predicting which team will win the Champions League in soccer

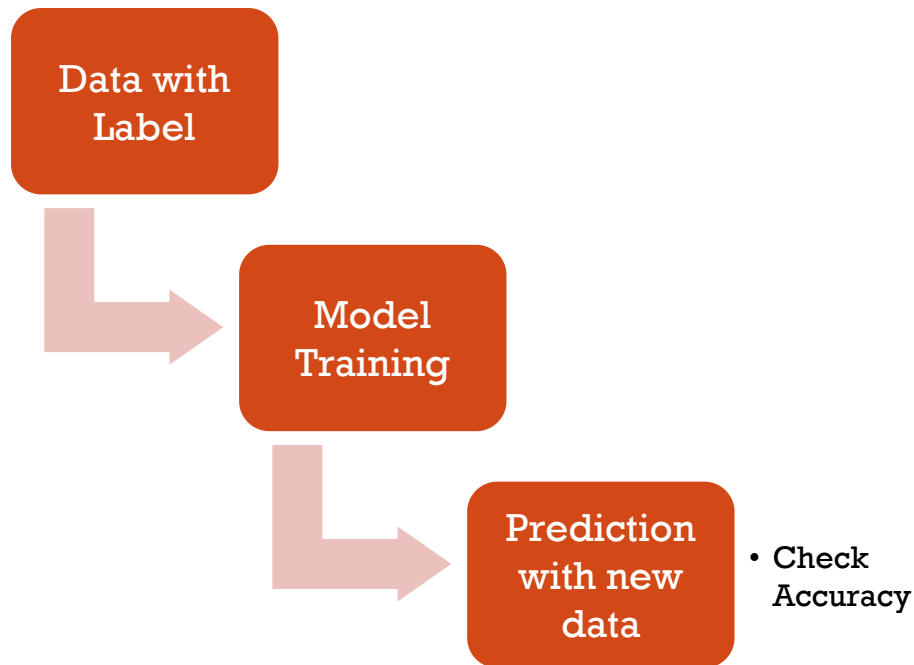


# TYPES OF MACHINE LEARNING

- **Supervised learning** techniques attempt to discern results and learn by trying to find patterns in a labeled data set. Human interaction is required to label the data.
- **Unsupervised learning** techniques don't rely on labeled data and attempt to find patterns in a data set without human interaction.
- **Semi-supervised learning** techniques need labeled data, and therefore human interaction, to find patterns in the data set, but they can still progress toward a result and learn even if passed unlabeled data as well.



# BASIC MACHINE LEARNING PIPELINE



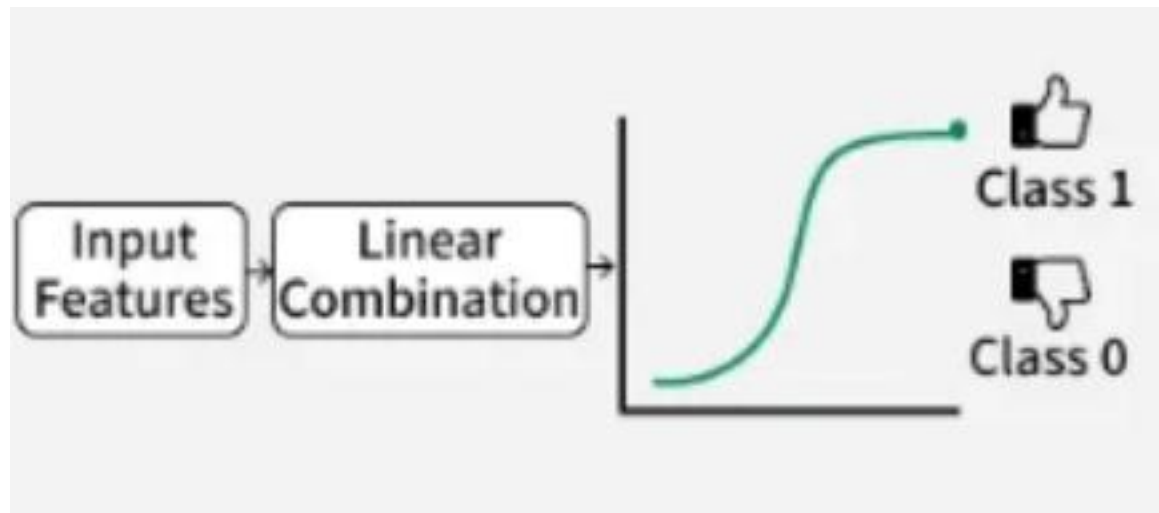
# SUPERVISED LEARNING( CLASSIFICATION)

- “Supervised classification is just teaching a computer how to sort things into categories, using examples where we already know the answers.”
  - Step 1: Training data – examples with labels (fruit with names, emails marked spam/not spam).
  - Step 2: Learning – computer finds rules/patterns.
  - Step 3: Prediction – apply rules to unseen data.
  - Step 4: Evaluation – check if the prediction is correct.



# LOGISTIC REGRESSION

- “a curve that separates two classes.”
- Logistic Regression takes that line and passes it through an S-shaped curve (sigmoid).
- The S-shaped sigmoid curve turns a line into probability.
- Why?
  - So predictions stay between 0 and 1 (like probabilities).



# HOW LOGISTIC REGRESSION WORKS??

- It transforms the linear regression function using a sigmoid function which maps any real-valued set of independent variables input into a value between 0 and 1. This function is known as the logistic function.
- Suppose we have input features represented as a matrix  $X$  and the dependent variable is  $Y$  having only binary value i.e 0 or 1.

$$X = \begin{bmatrix} x_{11} & \dots & x_{1m} \\ x_{21} & \dots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nm} \end{bmatrix}$$

$$Y = \begin{cases} 0 & \text{if Class 1} \\ 1 & \text{if Class 2} \end{cases}$$





# HOW LOGISTIC REGRESSION WORKS??

- Then, apply the multi-linear function to the input variables  $X$ .

$$z = \left( \sum_{i=1}^n w_i x_i \right) + b$$

- Here  $x_i$  is the  $i$ th observation of  $X$ ,  $w = [w_1, w_2, w_3, \dots, w_n]$  is the weights or Coefficient and  $b$  is the bias term also known as intercept.
- Simply this can be represented as the dot product of weight and bias.  $z = w \cdot X + b$



# LOGISTIC REGRESSION MODEL

- Logistic regression then applies the sigmoid function to  $z$  to convert it into a probability between 0 and 1 which can be used to predict the class.

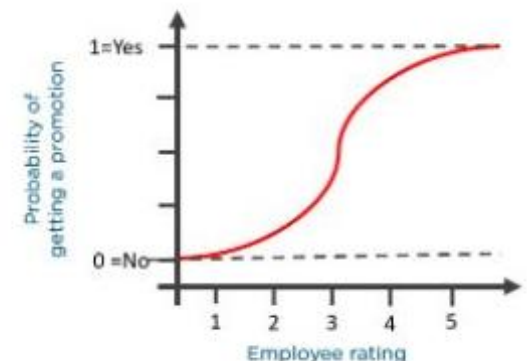
$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P(y=1 \mid x) = \sigma(z) = \frac{1}{1+e^{-z}}, \quad z = \beta_0 + \beta_1 x.$$



# EXAMPLE:

- Consider the following example: An organization wants to determine an employee's salary increase based on their performance.



# ODDS-RATIO

- To understand logistic regression, let's go over the odds of success.
  - Odds ( $\theta$ ) = Probability of an event happening / Probability of an event not happening
  - $\theta = p / 1 - p$
  - The values of odds range from zero to  $\infty$  and the values of probability lies between zero and one.
- Consider the equation of a straight line:  $y = \beta_0 + \beta_1 * x$
- Now to predict the odds of success, we use the following formula:

$$\log \left( \frac{p(x)}{1-p(x)} \right) = \beta_0 + \beta_1 x$$



- Exponentiating both the sides, we have:

$$\left( \frac{p(x)}{1 - p(x)} \right) = e^{\beta_0 + \beta_1 x}$$

$$\text{Let } Y = e^{\beta_0 + \beta_1 x}$$



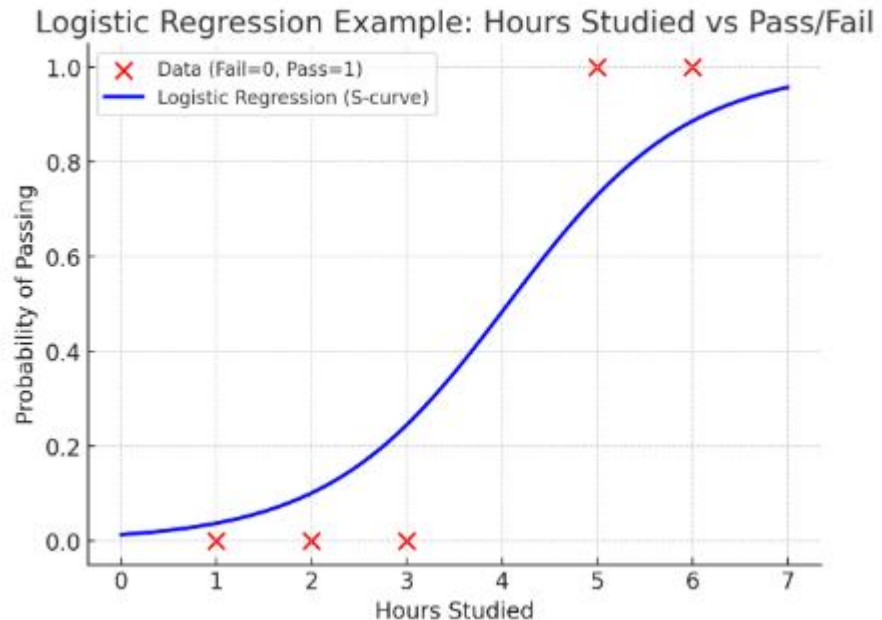
# EXAMPLE-LOGISTIC REGRESSION

## Objective:

Fit a logistic regression model that predicts whether a student passes (1) or fails (0) based on study hours.

“If a student studies 4 hours, what should we predict?”

| Hours Studied | Result |
|---------------|--------|
| 1             | Fail   |
| 2             | Fail   |
| 3             | Fail   |
| 5             | Pass   |
| 6             | Pass   |



$$P(y = 1|x) = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$$

$P(y = 1|x)$  = probability of "Pass".

$b_0$  = intercept (bias).

$b_1$  = slope (weight).

The exponential makes the curve **S-shaped** (sigmoid).

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|x) \geq 0.5 \\ 0 & \text{if } P(y = 1|x) < 0.5 \end{cases}$$



# EXAMPLE: LOGISTIC REGRESSION

We're working with this logistic regression setup:

$$p_i = \sigma(z_i), \quad z_i = \beta_0 + \beta_1 x_i, \quad r_i = p_i - y_i$$

and the Newton–Raphson update:

$$\Delta = (X^T W X)^{-1} X^T (p - y), \quad \beta^{(k+1)} = \beta^{(k)} - \Delta.$$

- Step# 1 : Compute initial probabilities and weights

$$z_i^{(0)} = \beta_0^{(0)} + \beta_1^{(0)} \cdot x_i \qquad p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}$$

$$z_i^{(0)} = 0 \quad \text{for all } i. \qquad p_i = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5$$





- The weights are based on how *uncertain* the predictions are.

$$w_i = p_i(1 - p_i)$$

Since  $p_i = 0.5$ :

$$w_i = 0.5 \times (1 - 0.5) = 0.25$$

| Student | Hours ( $x$ ) | Label ( $y$ ) | Linear predictor ( $z$ ) | Predicted prob ( $p$ ) | Weight ( $w$ ) |
|---------|---------------|---------------|--------------------------|------------------------|----------------|
| 1       | 1             | 0             | 0                        | 0.5                    | 0.25           |
| 2       | 2             | 0             | 0                        | 0.5                    | 0.25           |
| 3       | 3             | 0             | 0                        | 0.5                    | 0.25           |
| 4       | 5             | 1             | 0                        | 0.5                    | 0.25           |
| 5       | 6             | 1             | 0                        | 0.5                    | 0.25           |



Since  $w_i = 0.25$  for all, we can factor it out:

$$X^T W X = 0.25 \cdot X^T X.$$

Compute  $X^T X$ :

$$X^T X = \begin{bmatrix} \sum 1 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} = \begin{bmatrix} 5 & 17 \\ 17 & 75 \end{bmatrix}.$$

Multiply by 0.25:

$$X^T W X = \begin{bmatrix} 1.25 & 4.25 \\ 4.25 & 18.75 \end{bmatrix}.$$



Here  $p - y = r = [0.5, 0.5, 0.5, -0.5, -0.5]$ .

So:

$$X^T(p - y) = \begin{bmatrix} \sum r_i \\ \sum x_i r_i \end{bmatrix}.$$

- $\sum r_i = 0.5 + 0.5 + 0.5 - 0.5 - 0.5 = 0.$
- $\sum x_i r_i = 1 \cdot 0.5 + 2 \cdot 0.5 + 3 \cdot 0.5 + 5 \cdot (-0.5) + 6 \cdot (-0.5).$   
 $= 0.5 + 1 + 1.5 - 2.5 - 3 = -2.5.$

So:

$$X^T(p - y) = \begin{bmatrix} 0 \\ -2.5 \end{bmatrix}.$$

$$X^T W X = \begin{bmatrix} 1.25 & 4.25 \\ 4.25 & 18.75 \end{bmatrix}.$$

$$A^{-1} = \frac{1}{5.375} \begin{bmatrix} 18.75 & -4.25 \\ -4.25 & 1.25 \end{bmatrix}.$$



$$\Delta = (X^T W X)^{-1} X^T (p - y).$$

$$= \frac{1}{5.375} \begin{bmatrix} 18.75 & -4.25 \\ -4.25 & 1.25 \end{bmatrix} \begin{bmatrix} 0 \\ -2.5 \end{bmatrix}.$$

$$\Delta \approx \begin{bmatrix} 10.625 \cdot 0.1860 \\ -3.125 \cdot 0.1860 \end{bmatrix} = \begin{bmatrix} 1.9779 \\ -0.5814 \end{bmatrix}.$$

$$\beta^{(1)} = \beta^{(0)} - \Delta.$$

Since  $\beta^{(0)} = (0, 0)$ :

$$\beta^{(1)} \approx \begin{bmatrix} -1.9779 \\ 0.5814 \end{bmatrix}.$$



# DO THE NEXT ITERATION...

- $x = [1, 2, 3, 5, 6]$
- $y = [0, 0, 0, 1, 1]$
- $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}$
- Start:  $\beta^{(1)} = (\beta_0, \beta_1) = (-1.9779, 0.5814)$ .
- Sigmoid:  $\sigma(z) = 1/(1 + e^{-z})$ .

Step A — compute  $\int_{\mathbb{M}} J_{\mathbb{M}} \Gamma_{\mathbb{M}} \mathbb{B}_{\mathbb{M}}$  for each row



| i | x | y | $f_M$                  | $J_M = \mathcal{M}(f_M)$ | $r_M = J_M - \int_M$   | $B_M = J_M(1 - J_M)$  |
|---|---|---|------------------------|--------------------------|------------------------|-----------------------|
| 1 | 1 | 0 | $\frac{-1.396500}{00}$ | $\frac{0.198372}{10}$    | $\frac{0.198372}{10}$  | $\frac{0.159020}{61}$ |
| 2 | 2 | 0 | $\frac{-0.815100}{00}$ | $\frac{0.306804}{78}$    | $\frac{0.306804}{78}$  | $\frac{0.212675}{61}$ |
| 3 | 3 | 0 | $\frac{-0.233700}{00}$ | $\frac{0.441839}{47}$    | $\frac{0.441839}{47}$  | $\frac{0.246617}{35}$ |
| 4 | 5 | 1 | $\frac{0.929100}{00}$  | $\frac{0.716892}{66}$    | $\frac{-0.283107}{34}$ | $\frac{0.202957}{57}$ |
| 5 | 6 | 1 | $\frac{1.510500}{00}$  | $\frac{0.819135}{30}$    | $\frac{-0.180864}{70}$ | $\frac{0.148152}{66}$ |



## STEP B — BUILD THE PIECES FOR i <sup>بیا</sup> j <sup>بیا</sup> AND i <sup>بیا</sup> (J - I)

$$X^T W X = \begin{bmatrix} \sum_i w_i & \sum_i w_i x_i \\ \sum_i w_i x_i & \sum_i w_i x_i^2 \end{bmatrix}, \quad X^T (p - y) = \begin{bmatrix} \sum_i r_i \\ \sum_i x_i r_i \end{bmatrix}.$$

$$X^T W X \approx \begin{bmatrix} 0.96942381 & 3.22792773 \\ 3.22792773 & 13.63671445 \end{bmatrix}.$$



$$X^T W X \approx \begin{bmatrix} 0.96942381 & 3.22792773 \\ 3.22792773 & 13.63671445 \end{bmatrix}.$$

$$X^T(p - y) \approx \begin{bmatrix} 0.4830443 \\ -0.36322487 \end{bmatrix}.$$





## STEP C — COMPUTE THE NEWTON STEP $\Delta$

$$\Delta = (X^T W X)^{-1} X^T (p - y).$$

$$\Delta \approx \begin{bmatrix} 2.77105030 \\ -0.68256727 \end{bmatrix}.$$



# STEP D — UPDATE COEFFICIENTS

$$\beta^{(2)} = \beta^{(1)} - \Delta.$$

Numeric subtraction:

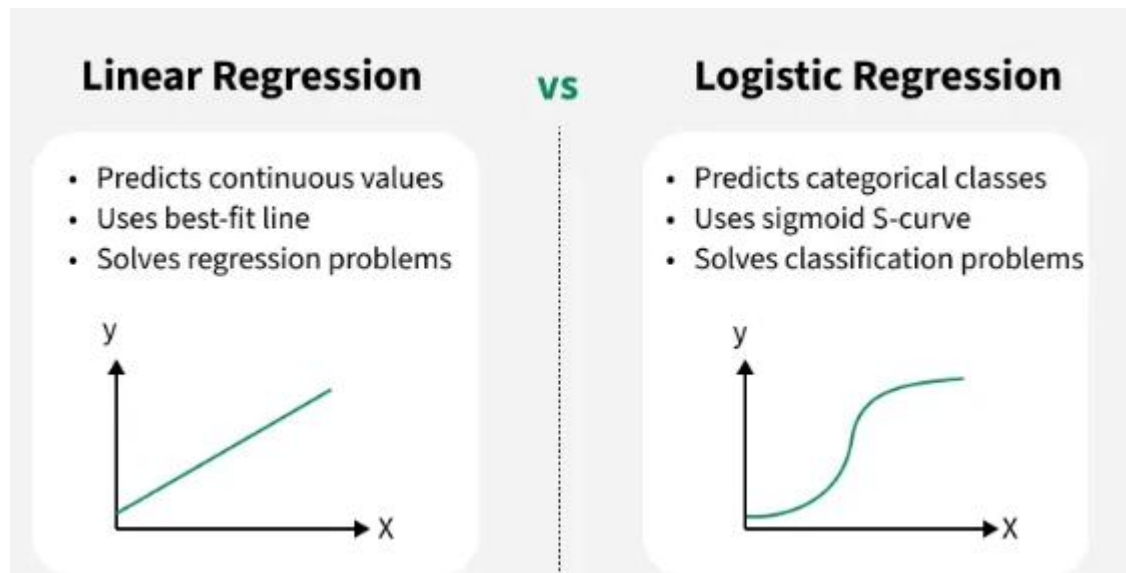
$$\beta^{(2)} \approx \begin{bmatrix} -1.9779 \\ 0.5814 \end{bmatrix} - \begin{bmatrix} 2.77105030 \\ -0.68256727 \end{bmatrix} = \begin{bmatrix} -4.74895030 \\ 1.26396727 \end{bmatrix}.$$

So after this **second** Newton/IRLS update we get

$$\beta^{(2)} \approx (-4.74895, 1.26397).$$



- Linear regression predicts any number.
- Logistic regression “squeezes” that into a probability (via sigmoid).
- Then applies a 0.5 cutoff to decide Pass/Fail.



# QUESTION:

- If the probability that a student passes an exam is modelled as

$$P(\text{Pass}) = \frac{1}{1 + e^{-(-3+0.2 \times \text{hours studied})}}$$

- Find the probability of passing if the student studied **10 hours**.
- At how many hours studied is the probability of passing **50%**?

